Research Paper Findings

Project 1

October 30th, 2026

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1 Trey Trejo - Paper Summary

1.1 Paper Name

Student Academic Success Prediction Using Learning Management Multimedia Data With Convoluted Features and Ensemble Model, by Abdullah Al-Ameri, Waleed Al-Shammari, Aniello Castiglione, Michele Nappi, Chiara Pero, and Muhammad Umer. ACM Journal of Data and Information Quality, volume 17, issue 3, 2025 (special issue on multimedia big data quality). DOI: 10.1145/3687268.

https://dl.acm.org/doi/10.1145/3687268

1.2 Contribution

The paper proposes a pipeline for predicting academic success from learning-management-system data. A custom one-dimensional convolutional neural network (CNN) turns each student's tabular record into a long vector of "convoluted" features, which are then used to train conventional classifiers. The authors claim this removes the need for manual feature engineering and beats using machine learning or deep learning models on their own. They state three objectives which introduce an RF–SVM (Random Forest and Support Vector Machine) ensemble built on the convoluted features, measure the effect of convoluted features by comparing models on original and convoluted features, and benchmark nine classifiers and the final ensemble against published studies. The text is framed around multimedia big-data quality in education.

1.3 Data Points Used

The dataset combines student activity from Moodle (an LMS similar to Canvas), video interaction from eDify (a cloud platform that stores student data), and academic information from a student information system. After mapping, it contains 326 instances. Section 3.1 lists 21 features: CGPA, Prohibition, Remote Student, High Risk, Likes, Term Exceeded, Other Modules, Plagiarism history, CW1, CW2, Applicant Name, At Risk, At Risk SSC, ESE, Online c, Online O, Played, Paused, Attempt Count, Segment, and Result. Categorical values are label-encoded. The target is described as grade categories "such as G1, G2, G3". The data is split 70% training and 30% testing (about 228 and 98 students), and a separate 10-fold cross-validation is also reported. It is not stated whether the dataset comes from one course, one institution, or one term.

1.4 Results

Table 2 reports single-model accuracy on original versus convoluted features. On original features the Random Forest is the best single model (84.31%), followed by the Extra Trees classifier (82.56%) and SVM (81.76%), with KNN (K-Nearest Neighbors) lowest (74.77%). On convoluted features Extra Trees (86.33%) and Random Forest (86.32%) are basically tied, followed by SVM (85.18%), with KNN again lowest (79.54%). Every model improves with convoluted features, by between about 2 and 5.5 points.

For pairwise soft-voting ensembles, the RF–SVM pair is best in both cases. On original features it reaches 91.35% accuracy (precision 91.28, recall 94.54, F1 92.40). On convoluted features it reaches 97.88% accuracy, with precision, recall, and F1 listed as 98.88 in Table 4. Table 6 compares the proposed model against twelve published studies, with the proposed row listed at 98.99% accuracy.

1.5 Method

Nine scikit-learn classifiers are first trained on the original features: Random Forest, Extra Trees, Gradient Boosting, Decision Tree, KNN, Logistic Regression, Gaussian Naive Bayes, linear-kernel SVM, and SGD (Stochastic Gradient Descent). Hyperparameters are listed in Table 1 (for example, Random Forest with 200 trees and maximum depth 20; SVM with linear kernel and $C = 2.0$), and the paper says they were chosen by grid search. A CNN is then applied: an embedding layer (vocabulary size 20,000, 300 output dimensions), a 1D convolution with 5,000 filters and ReLU (Rectified Linear Unit), max-pooling of size 2, and a flatten layer. The nine models are retrained on the flattened output. The best individual models are combined in pairs by voting, and the RF–SVM pair is selected. Evaluation uses accuracy, precision, recall, and F1.

1.6 Limitations

Section 4.6 gives two. First, the RF–SVM ensemble with CNN feature extraction is computationally expensive and may not suit institutions with limited infrastructure. Second, convoluted features may not generalize to unseen data, depend heavily on data quality, and may miss factors such as socioeconomic background, motivation, and learning environment. The conclusion adds that future work will try ensembles of machine learning and deep learning models and other datasets.

Very small data, with 326 instances, the 30% test set holds about 98 students, so each student is worth roughly one point of accuracy. The paper uses a random 70/30 split and random cross-validation, with no second dataset, no confidence intervals, and no significance testing. The authors list extending to other datasets as future work.

The paper describes extracting CNN features from the dataset and then training the models, followed by the 70/30 partition. It does not state whether the CNN is trained, on what target, or whether it was fit before the split. The architecture also produces a feature vector with thousands of dimensions per student (5,000 filters).

Convoluted features are not tied to named inputs, so per-student explanations are not available, and no subgroup analysis appears despite the dataset containing a remote-student indicator and other attributes. The paper also does not discuss student privacy.

1.7 How the Paper Applies to Our Project

Our project predicts non-re-enrollment potentially using a Random Forest, an observed label, fixed as-of dates, and a leakage test. The paper is useful as a source of a few potential ideas.

1.8 Table of Paper Elements

Paper element	Relevance to our project	Potential action
Random Forest as a strong single model	On the paper's data, RF is the best single model on original features (Table 2) and tied for best on convoluted features. This weakly supports RF as a sensible default.	Keep RF as the model; do not treat 326 random-split rows as strong evidence

Paper element	Relevance to our project	Potential action
RF hyperparameters (200 trees, depth 20)	A plausible starting point but not shown to be tuned or validated	Use only as one point in a search, tuned with time-aware validation and recorded in the registry
Feature list with outcome-like columns and Applicant Name	Shows how outcome-derived and identifier columns can end up among inputs	Use a feature allowlist, not a denylist, and fail the build on any unlisted column
CNN convoluted features	Thousands of derived features for hundreds of rows and no explanation path. Out of scope this phase.	Initially defer, if tried later we need to come up with adequate feature safeguards
Nine-model comparison on two feature sets	A good structure to isolate and group performance	Reuse the structure (same models, same split, features swapped) in a later phase, not the results
Extra Trees and linear SVM	Cheap, competitive comparison points in the paper's own tables	Add both as baselines
Data-quality framing	Emphasis on periodic data capture snapshots, label rules, and leakage tests.	Include the use of periodically capturing data to continuously improve the Random Forest

1.9 What Can Be Transfered To Our Project

The paper offers three takeaways. First, Random Forest and Extra Trees are strong, inexpensive options on small tabular student data. Second, a controlled comparison of the same models on two feature sets under a single protocol is a good design to isolate group performance. Third, the paper's limitation about computation is a fair reminder to record training cost and to test on later cohorts and other populations.

References

[1] A. Al-Ameri, W. Al-Shammari, A. Castiglione, M. Nappi, C. Pero, and M. Umer. Student Academic Success Prediction Using Learning Management Multimedia Data With Convoluted Features and Ensemble Model. *ACM Journal of Data and Information Quality*, 17(3), 2025. DOI: 10.1145/3687268.

Contents

2 Jackson Marcum – Paper 1 Summary

2.1 Paper Name

A Modular and Explainable Machine Learning Pipeline for Student Dropout Prediction in Higher Education. *Algorithms*, volume 18, issue 10, article 662 (2025).
https://www.mdpi.com/1999-4893/18/10/662

2.2 Contribution

This paper looks at how different machine learning models can predict student dropouts and how universities can understand those predictions. The researchers built a pipeline that compares nine classifiers and then uses SHAP to explain which student features are affecting the predictions. One useful part is that the system does not just give a risk score. It can also flag uncertain or unusual predictions for a person to review. This matters for our project because a prediction is more useful if we can explain it and check that it is reliable.

2.3 Data Points Used

The main dataset contains 17,883 undergraduate records from a Moroccan higher education institution after cleaning, with 91 features. Of these students, 2,856 (about 16%) were labeled as dropouts. The data includes academic performance and administrative information, such as GPA, passed courses, earned credits, and academic or enrollment dates. The outcome was taken from the registrar's student-status field at the end of the observation window. The researchers did not use that outcome field as an input feature. They used a stratified 80/20 split, with 14,306 students for training and 3,577 for testing. They also repeated their comparison on a separate public Portuguese student dropout dataset.

2.4 Results

On the main test set, XGBoost had the strongest overall results, with 97.0% accuracy, 0.993 AUC-ROC, 0.911 F1-score, and 94.4% recall for dropouts. Random Forest was also competitive, with 96.8% accuracy, 0.989 AUC-ROC, 0.898 F1-score, and 87.7% recall. However, the difference between XGBoost and Random Forest was not statistically significant in the authors' paired error test ($p = 0.428$). On the external dataset, XGBoost's F1-score dropped to 0.792, which shows that performance can change on another institution's data. The SHAP analysis found that academic timing, number of passed courses, GPA, and earned credits were among the important prediction features.

2.5 Method

The researchers tested logistic regression, decision tree, Random Forest, k-nearest neighbors, support vector machine, gradient boosting, XGBoost, Naive Bayes, and a multilayer perceptron. They used five-fold cross-validation on the training data and then evaluated the final models on a held-out test set. They reported accuracy, precision, recall, F1-score, and AUC-ROC. SHAP was used after training to show which features pushed an individual prediction toward dropout or retention. They also compared model errors statistically and examined performance across academic subgroups. This is more detailed than simply choosing whichever model has the highest accuracy.

2.6 Limitations

One limitation is that the main data came from one institution, so its performance may not transfer directly to our student population. A second limitation is that the results were lower on the external dataset, showing that the original scores are not guaranteed elsewhere. Third, some important features are administrative dates and status-related information, so our project would need to check that these values are actually available before the prediction date and do not leak the outcome. Fourth, the fairness analysis only used academic groups such as major and credit load, not demographic groups that were unavailable or not approved for analysis. Fifth, SHAP explains what affected a prediction but does not prove those features

caused a student to drop out. Another limitation is that the model used mostly tabular records rather than tracking how student behavior changes over time. These are a mix of limitations the authors discuss and concerns relevant to applying the study to our project.

2.7 How the Paper Applies to Our Project

Our group is looking at predicting whether students will re-enroll, potentially using Random Forest and data collected before a fixed prediction date. This paper supports keeping Random Forest as a strong baseline, but it also gives us a reason to compare it against XGBoost later. More importantly, it shows why we should record the prediction date, avoid outcome-related inputs, report dropout recall, and explain why a student was flagged. We should not assume the paper's high accuracy will carry over to our dataset.

2.8 Table of Paper Elements

Paper element	Relevance to our project	Potential action
Random Forest baseline	Strong performance close to XGBoost	Start with RF and measure recall, F1, and precision
XGBoost comparison	Higher dropout recall on the main test set	Consider as a later benchmark
SHAP explanations	Helps explain individual risk predictions	Add feature explanations if time allows
Academic and administrative features	GPA and credits may be useful, but dates can leak outcomes	Make a feature allowlist and check availability at prediction time
External validation	Results changed on a second dataset	Test on later student cohorts if possible
Subgroup evaluation	Average performance may hide uneven errors	Compare results across allowed academic groups

2.9 What Can Be Transferred to Our Project

The three main takeaways are to keep Random Forest as a reasonable starting model, evaluate how well we actually catch students who do not re-enroll, and make sure every input feature would have been known when the prediction was made. The SHAP approach could also help us explain predictions if we have time to add it. I would treat the paper's results as a useful comparison rather than a target accuracy that our model must reach.

3 Jackson Marcum – Paper 2 Summary

3.1 Paper Name

Predictive analytics study to determine undergraduate students at risk of dropout, by Andres Gonzalez-Nucamendi, Julieta Noguez, Luis Neri, Victor Robledo-Rella, and Rosa Maria Guadalupe Garcia-Castelan. *Frontiers in Education*, volume 8 (2023). DOI: 10.3389/feduc.2023.1244686.
https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2023.1244686/full

3.2 Contribution

This study focuses on identifying undergraduate students who might drop out early enough for universities to offer support. The researchers compared machine learning classifiers and looked at which academic factors were most useful. What makes this paper important is that it focuses on a problem with dropout data: most students stay enrolled, so a model can look accurate while missing the students who actually drop out. The researchers changed the classification probability threshold to improve the balance between finding dropouts and correctly identifying students who stay.

3.3 Data Points Used

The study used records from 14,495 undergraduate students at Tecnologico de Monterrey in Mexico. About 8.5% dropped out and 91.5% were retained. The researchers looked at early first-semester academic performance, previous average grades, mathematics scores, university entrance exam scores, number of class hours, age, scholarship funding, English level, and subjects dropped during the early weeks. This means the data includes both information known before college and information collected shortly after classes begin. That timing is important if we want our own model to predict non-re-enrollment before it happens.

3.4 Results

Using the standard 50% classification threshold, the study reported only 13.2% correct classification for dropout cases while correctly classifying 99.4% of retention cases. This shows how much the majority class affected the predictions. Random Forest was selected from eight classifiers because it gave a useful balance and needed less tuning. By adjusting the threshold, the researchers increased correct dropout classification above 50% while keeping retention and overall classification above 70%. The important variables included early grades, previous grades, mathematics scores, and entrance exam results. The main takeaway is that overall accuracy by itself can be misleading when dropouts are uncommon.

3.5 Method

The researchers considered eight classifiers: support vector machine, k-nearest neighbors, decision tree, Random Forest, AdaBoost, XGBoost, Naive Bayes, and linear discriminant analysis. They also examined student groups and the predictive strength of individual variables. Random Forest was selected for further analysis. Normally, a student would be assigned to the dropout class if the predicted probability crossed a 0.5 threshold. The researchers adjusted that cutoff to catch more of the minority dropout group. They compared the effect on correct dropout classification, correct retention classification, and overall performance. The paper uses the term "precision" in its own reporting; when using these results in our project, we should be clear about exactly which class-based metric is being reported.

3.6 Limitations

One limitation is the very unbalanced dataset, since only 8.5% of students dropped out. A second limitation is that improving dropout detection by changing the threshold can increase false alarms for students who would stay. Third, the study is based on one university in Mexico, so its patterns may not be the same at other schools. Fourth, several strong predictors come from the first weeks of classes, which means those values cannot be used for predictions made before the semester begins. Fifth, the study mainly uses

recorded academic and background information, so personal circumstances that affect dropout may be missing. Another limitation for our project is that the paper predicts dropout, while our group is focused on non-re-enrollment, and those outcomes are not necessarily defined the same way. These points include both study limitations and concerns about transferring its methods to our task.

3.7 How the Paper Applies to Our Project

This paper is especially relevant if our group uses Random Forest. It shows why we cannot judge the model only by its overall accuracy. If most students re-enroll, a model that predicts everyone will return could still look accurate while failing at the main goal. We should measure how many actual non-re-enrolling students we identify and how many returning students we incorrectly flag. The study also gives us an idea for testing different probability thresholds, but that threshold should be chosen using validation data rather than changed after seeing the test results.

3.8 Table of Paper Elements

Paper element	Relevance to our project	Potential action
Random Forest	Selected as a practical classifier	Use RF as a starting model
Class imbalance	Dropouts were only 8.5% of records	Report class counts and a majority-class baseline
Threshold adjustment	Changes the balance of missed students and false alarms	Tune threshold on validation data
Early academic features	Grades and dropped classes can predict risk	Only use them after they exist at the snapshot date
Previous grades and entrance scores	May provide useful early signals	Check which comparable fields are available
Dropout versus non-re-enrollment	Different definitions can change labels	Write a clear target-label rule

3.9 What Can Be Transferred to Our Project

The biggest takeaway is that the model needs to find students who do not return, not just get a high overall accuracy. We can transfer the idea of evaluating both classes separately and trying a probability threshold that fits our project's goal. We can also look at previous academic performance as possible features, while being careful about when those features become available. Before training, we should define exactly what counts as non-re-enrollment and keep that outcome separate from our inputs.

References

1. *A Modular and Explainable Machine Learning Pipeline for Student Dropout Prediction in Higher Education. Algorithms* 18(10), 662 (2025). https://www.mdpi.com/1999-4893/18/10/662.
2. A. Gonzalez-Nucamendi, J. Noguez, L. Neri, V. Robledo-Rella, and R. M. G. Garcia-Castelan. *Predictive analytics study to determine undergraduate students at risk of dropout. Frontiers in Education* 8 (2023). https://doi.org/10.3389/feduc.2023.1244686.

4 Owen Badzinski — Paper Summary

4.1 Paper Name

Predicting and Mitigating Freshmen Student Attrition: A Local-Explainable Machine Learning Framework, by Dursun Delen, Behrooz Davazdahemami, and Elham Rasouli Dezfouli. Published in *Information Systems*

Frontiers, April 2023. DOI: 10.1007/s10796-023-10397-3.

https://pmc.ncbi.nlm.nih.gov/articles/PMC10097523/

4.2 Contribution

The paper proposes a hybrid machine learning framework that combines student attrition prediction with explainable AI (SHAP). Its primary contribution is demonstrating how individual-level explanations can help universities understand why specific students are at risk of dropping out, rather than relying exclusively on general feature importance across the entire student population. The authors argue that different students can have similar predicted dropout probabilities but require different interventions because their underlying risk factors differ. The study also incorporates socioeconomic information, including state-level GDP, unemployment rates, and per-capita income, alongside university enrollment records. The goal is to help universities identify at-risk students and provide more appropriate, individualized interventions.

4.3 Data Points Used

The dataset combines freshman enrollment records from a Midwestern U.S. university with socioeconomic information from three government sources. It contains 39,470 unique students covering 10 years (2009–2018). The university dataset originally contained more than 60 features, which were reduced to 34 selected input features.

The data includes demographic information (age, gender, race, first-generation status, and in-state/out-of-state status), pre-college educational information (high school GPA, ACT scores, years after high school, and transfer credits), college performance (fall GPA, credits attempted, credits earned, and pass ratio), college life (university housing and Greek life participation), and financial information (financial aid, grants, loans, and work-study).

Three additional socioeconomic features were collected: state-level GDP from the Bureau of Economic Analysis, per-capita income from the U.S. Census Bureau, and unemployment rates from the Bureau of Labor Statistics. These were matched to students based on their state of residence and enrollment year.

The target variable is whether a student enrolled in the fall semester of their sophomore year. Approximately 79.6% of students were retained, while 20.4% experienced attrition.

4.4 Results

The best-performing model achieved 88.4% overall accuracy and an ROC-AUC of 0.886. The model achieved 77.4% recall for students who dropped out and 91.1% specificity for students who returned.

The researchers found that academic performance was particularly important in predicting student attrition. Three of the five most important features were related to students' ability to pass courses and maintain satisfactory grades. Students who did not enroll in the spring semester of their freshman year had a particularly high attrition rate, with approximately 94% never returning for the following fall. Additionally, more than 89% of students who participated in Greek life returned for their sophomore year.

The study also found that socioeconomic information improved predictive performance. When GDP, unemployment rate, and per-capita income were removed, accuracy decreased from 88.4% to 84.4%, while attrition recall decreased from 77.4% to 72.4%. These differences were statistically significant.

An important finding from the SHAP analysis was that two students with nearly identical predicted dropout probabilities (81.9% and 81.2%) had very different contributing factors. One student's risk was primarily associated with academic performance, while the other student's circumstances suggested potential financial difficulties. This demonstrates the value of individual-level explanations when determining appropriate student interventions.

4.5 Method

The researchers developed a fully connected Multilayer Perceptron (MLP) deep neural network to predict student attrition. The model contained five hidden layers, used ReLU activation functions in the hidden layers, and a sigmoid activation function in the output layer. After preprocessing the 34 original features through binning and one-hot encoding, the dataset contained 150 input variables.

The researchers used grid search to optimize hyperparameters, testing 1,800 combinations involving the optimizer, learning rate, decay rate, batch size, and number of training epochs. Weighted binary cross-entropy was used to address class imbalance because only 20.4% of the students experienced attrition. Learning-rate decay was also used to help prevent overfitting.

For explainability, the researchers used SHAP (SHapley Additive exPlanations) through the KNIME Analytics Platform. SHAP was used to calculate individual feature contributions for each student's predicted attrition probability. The researchers also calculated average absolute SHAP values across the dataset to identify the most important features at the population level.

The methodology therefore provides both global explanations (important factors across the student population) and local explanations (factors contributing to an individual student's prediction).

4.6 Limitations

The authors identify several limitations. First, the study only examines freshman students from a single U.S. university. The results may not generalize to institutions with different student populations, policies, and environments. The authors recommend testing the framework across multiple universities and examining attrition during later college years.

Second, the dataset only includes students who attended traditional in-person instruction before the COVID-19 pandemic. The findings may not generalize to modern online and hybrid learning environments.

Third, the dataset does not include psychological and cultural factors that may influence student attrition. The authors suggest incorporating additional information from surveys, advising records, and counseling services in future research.

Finally, the paper does not experimentally evaluate whether the individualized interventions suggested by SHAP actually improve student retention. The authors recommend longitudinal research to determine whether these interventions are more effective than existing approaches.

Another consideration for our project is that the study uses deep learning rather than traditional machine learning. The five-layer neural network and extensive hyperparameter search introduce additional computational complexity that may not be necessary for our project's objectives.

4.7 How the Paper Applies to Our Project

Our project predicts student outcomes using historical university data, potentially including non-re-enrollment as the target variable. The paper is particularly useful for its approach to explainability and its use of multiple categories of student information.

The SHAP methodology could be applied to our Random Forest model to explain why individual students receive higher risk predictions. Instead of only displaying a student's predicted risk, our platform could identify which academic, financial, or engagement-related features contribute most to that prediction. This would provide academic advisors with more useful information when determining how to support students.

The paper also provides a useful example of combining demographic, educational, financial, and socioeconomic information. We could use these categories to organize our available institutional data and investigate which features provide useful predictive information.

Additionally, the researchers' use of 10 years of historical student records highlights the importance of evaluating model performance across different entering classes. Our proposed rolling-origin validation approach could help determine whether a model trained on earlier cohorts generalizes to later cohorts.

The paper does not directly establish that SHAP recommendations improve retention, but it provides a framework for making model predictions more interpretable and useful for academic advising.

4.8 Table of Paper Elements

Paper element	**Relevance to our project**	**Potential action**
SHAP explainability	Demonstrates how individual student predictions can be explained using contributing features.	Implement SHAP with our Random Forest model to provide individual student explanations.
Global feature importance	Identifies the most influential factors across the entire student population.	Use aggregated SHAP values to identify common academic-risk factors.
Individual feature importance	Shows that students with similar risk predictions can have different underlying circumstances.	Display individual contributing factors in the advisor dashboard.
Multiple data categories	Combines demographic, educational, financial, college-life, and socioeconomic information.	Organize available institutional data into similar categories and evaluate their usefulness.
Class imbalance	Only 20.4% of students experienced attrition, requiring cost-sensitive learning.	Evaluate class imbalance in our dataset and consider class weighting or other appropriate approaches.
Academic performance features	GPA, credit completion, and pass ratio were among the most influential features.	Investigate GPA, attempted credits, earned credits, and course performance in our EDA.
Socioeconomic features	External economic indicators improved the model's predictive performance.	Investigate whether relevant socioeconomic data is available and appropriate for our project.
Historical cohorts	Uses 10 years of student records and examines differences between entering classes.	Use time-based validation to evaluate performance across historical entering classes.
Individualized interventions	Connects individual risk factors to potential student support services.	Design dashboard explanations that help advisors investigate potential support options without automatically prescribing interventions.
MLP neural network	Achieved 88.4% accuracy but requires extensive hyperparameter optimization.	Do not directly adopt the deep learning model; compare traditional ML approaches instead.

4.9 What Can Be Transferred To Our Project

The paper offers three main takeaways. First, SHAP can provide both global and individual explanations, making it useful for understanding why students receive particular risk predictions. This is especially relevant to our project because our platform aims to provide useful information to academic advisors rather than simply classify students as at-risk or not at-risk.

Second, combining multiple categories of student information can improve predictive performance and provide a more complete understanding of student attrition. We should investigate which academic, financial, demographic, and engagement features are available in our institutional datasets and determine whether they contribute useful predictive information.

Third, evaluating student outcomes across different cohorts is important because student populations and institutional conditions can change over time. Our rolling-origin validation approach could help evaluate whether the model generalizes across entering classes.

The paper's five-layer MLP does not need to be transferred directly. SHAP can also be used with traditional machine learning algorithms, including Random Forest and gradient boosting. Therefore, we can adopt the paper's explainability framework while maintaining our project's focus on non-deep-learning methods.

Overall, this paper is most useful as a reference for our explainability approach, feature analysis, and advisor-facing decision-support functionality.

References

[1] D. Delen, B. Davazdahemami, and E. Rasouli Dezfouli. Predicting and Mitigating Freshmen Student Attrition: A Local-Explainable Machine Learning Framework. *Information Systems Frontiers*, 2023. DOI: 10.1007/s10796-023-10397-3. https://pmc.ncbi.nlm.nih.gov/articles/PMC10097523/

5 Kellen Swanson — Paper Summary

5.1 Paper Name

The Use of Semester Course Data for Machine Learning Prediction of College Dropout Rates, by Viktor Kiss, Edgar Maldonado, and Mark Segall. Published in *Journal of Higher Education Theory and Practice*, Vol. 22, No. 4, 2022, pp. 64–74.

https://search.proquest.com/openview/b1527390c039f14984497f4472bbe95f/1?

5.2 Contribution

The paper investigates how granular semester-wide course performance data and student demographics can be leveraged to predict student dropout rates using supervised machine learning algorithms. Its primary contribution is demonstrating that traditional, interpretable machine learning models like logistic regression can outperform more complex architectures (such as neural networks and support vector machines) when utilizing semester course enrollment and performance records. The study provides empirical evidence on how tracking course-level progression across multiple semesters allows institutions to build effective early warning systems for student retention.

5.3 Data Points Used

The dataset comprises institutional student records covering active semesters ranging from Fall 2007 to Fall 2019, encompassing 21,079 unique students. The study incorporates student demographic information (including age, gender, and race/ethnicity, noting that approximately 20% of students started after their 30th birthday, 59% were male, and roughly 30% were of Hispanic, African American, or Asian descent) alongside comprehensive course enrollment and performance history.

To ensure data quality during preprocessing, courses taken by fewer than 100 students were removed, retaining 276 unique courses in the final analysis. The data was split into 70% training, 15% validation, and 15% testing sets. The target binary variable distinguishes between students who graduated/retained versus those who experienced dropout.

5.4 Results

Among the four evaluated machine learning techniques, Logistic Regression achieved the highest overall predictive accuracy at 84.8%. This was followed by Decision Trees at 82.2%, Neural Networks at 80.8%, and Support Vector Machines at 72.5%.

These results are benchmarked against a null error rate of 42.4% (where a baseline predicting the majority dropout class of 57.6% would be incorrect 42.4% of the time). The findings highlight that simpler, highly interpretable models can yield superior performance and practical utility over complex deep learning structures when working with semester-level academic tracking data.

5.5 Method

The research frames student dropout prediction as a binary classification problem. Four candidate machine learning algorithms were implemented and evaluated using Python's Scikit-learn library:

1. **Logistic Regression:** Modeled linear relationships between semester performance metrics, demographics, and dropout status, achieving top accuracy.
2. **Decision Trees:** Iteratively partitioned data based on feature splits to capture non-linear decision boundaries.
3. **Neural Networks:** Utilized multi-layer perceptron architectures to model complex interactions among features.
4. **Support Vector Machines (SVM):** Constructed hyperplanes to separate student outcome classes in feature space.

Model hyperparameters were tuned using the 15% validation dataset, and final performance metrics were reported from evaluation on the held-out 15% test dataset.

5.6 Limitations

The authors note several limitations. First, the study relies on data from a single institution, which may limit the generalizability of the exact feature thresholds and model weights to universities with different demographic compositions, academic policies, or grading structures.

Second, the dataset primarily reflects traditional enrollment dynamics across historical cohorts up to 2019, meaning the patterns might not fully account for post-pandemic shifts in remote learning, hybrid course formats, or changing economic stressors.

Finally, the study does not deeply incorporate qualitative, psychological, or real-time advising interaction data, which could provide additional explanatory context for student disengagement.

5.7 How the Paper Applies to Our Project

This paper directly informs our project's approach to predicting student academic outcomes and non-re-enrollment risks. By demonstrating that Logistic Regression (84.8%) and Decision Trees (82.2%) outperform or match complex neural networks on tabular student data, the paper validates our preference for interpretable, traditional machine learning models over overly complex deep learning pipelines.

Furthermore, the paper's emphasis on semester-wide course performance data highlights the importance of incorporating granular term-by-term academic progress metrics rather than relying solely on aggregate entry statistics. This supports our feature engineering strategy of tracking term GPA, credit accumulation rates, and specific course hurdles.

5.8 Table of Paper Elements

Paper element	Relevance to our project	Potential action
Semester course data	Demonstrates the predictive power of term-by-term academic performance records.	Incorporate semester-level credit completion and GPA trends into our feature matrix.
Logistic regression benchmark	Shows that simpler models can outperform neural networks on tabular academic records.	Include logistic regression alongside random forest as a baseline model.
Train/validation/test split	Provides a robust 70/15/15 data partitioning methodology.	Adopt rigorous data splitting and validation protocols in our model pipeline.
Course filtering preprocessing	Filters out low-enrollment courses (¡100 students) to reduce noise.	Apply similar sparsity thresholds during feature selection for course-level variables.
Binary classification target	Frames dropout/retention prediction as a clear binary classification task.	Align our project's target variable definition with standard retention markers.
Demographic integration	Combines age, gender, and ethnicity data with academic indicators.	Ensure demographic features are appropriately encoded and evaluated for equity.
Null error rate comparison	Uses a 42.4% null error benchmark to evaluate model skill.	Establish clear baseline error benchmarks for our predictive models.
Model interpretability	Emphasizes transparent models that advisors can easily interpret.	Prioritize model transparency to support advisor decision-making workflows.
Institutional generalizability	Highlights limits of single-institution historical datasets.	Account for cohort-specific shifts when validating model performance across years.
Feature granularity	Captures detailed academic history across multiple active semesters.	Aggregate longitudinal student metrics across multiple academic terms.

5.9 What Can Be Transferred To Our Project

The study offers three primary takeaways for our project. First, complex deep learning models are not strictly necessary for accurate student retention prediction; traditional, interpretable models like logistic regression and decision trees can achieve superior or comparable accuracy while remaining much easier for academic advisors to understand.

Second, utilizing structured semester-by-semester academic performance tracking provides a robust signal for early warning systems. Rather than relying exclusively on admissions data, capturing term-by-term progress allows models to detect shifting risk trajectories as students advance through their degree programs.

Third, clean data preprocessing—such as filtering out sparse courses with minimal enrollment—helps reduce noise and improves model stability. We can directly apply these data cleaning and validation principles to our own institutional dataset pipelines.

References

[1] V. Kiss, E. Maldonado, and M. Segall. The Use of Semester Course Data for Machine Learning Prediction of College Dropout Rates. *Journal of Higher Education Theory and Practice*, 22(4):64–74, 2022.